DECK 13 · WEEK 4 · MORNING 1

# Fixed Income I
# Instruments, Markets & Issuance

*How issuers finance themselves and investors get paid — from indentures to auctions*

CFA Level I · Vol 6 Learning Modules 1–5

# Roadmap — Fixed Income I — Instruments, Markets & Issuance

*5 sections · ~2 hours · Built bottom-up from CFA readings*

**01**

### Bond features & indenture

Six bond features · fixed / floating / zero coupons · FRN math · affirmative vs negative covenants · callable bonds (VIVU schedule)

**02**

### Cash-flow structures

Bullet vs amortizing vs balloon · USD 400k 30y mortgage · sinking funds · waterfalls · FRN / step-up / PIK / TIPS · puts & convertibles (ZTGB)

**03**

### Issuance & trading mechanics

Credit × maturity segmentation · S&P ratings · bond indexes · IG corporate issuance timeline · OTC secondary markets

**04**

### Corporate FI markets

Lines of credit · commercial paper & ABCP · repo mechanics + haircut · reverse-repo short-selling · Bear Stearns · IG vs HY

**05**

### Government & supranational issuers

Economic balance sheet · Ricardian equivalence · single- vs multiple-price auctions · agencies · supranationals

# Learning objectives

*By the end of this deck you should be able to:*

1. Describe the six features of a fixed-income security (issuer, maturity, principal, coupon, seniority, contingency provisions) and reproduce them from a prospectus summary.

2. Distinguish bullet, amortizing, and balloon cash-flow structures; compute the periodic payment of a fully amortizing mortgage and decompose each payment into interest and principal.

3. Contrast affirmative and negative covenants; identify the incremental covenant stack imposed on high-yield versus investment-grade issuers.

4. Compute the purchase price, repurchase price, and haircut of a repurchase agreement given an initial margin, and explain why repos are a short-term funding lifeline and a systemic vulnerability.

5. Compare single-price and multiple-price sovereign auctions; explain the role of primary dealers and the "non-economic" demand that compresses sovereign yields below corporate curves.

SECTION 01

# Bond features & indenture

# Every bond is fully specified by six features — issuer, maturity, principal, coupon, seniority, and contingency provisions — spelled out in the prospectus

*Worked reference — BRWA Corporation 3.2% Five-Year Notes, USD 300m par, senior unsecured, semi-annual coupon → USD 4.8m every six months.*

| Feature | What it defines | BRWA 3.2% Five-Year Notes |
|---|---|---|
| Issuer | Legal entity liable for interest and principal | Bright Wheels Automotive Corp. |
| Maturity | Date of final payment; tenor = remaining life | 5 years from settlement |
| Principal (par) | Amount repaid at maturity | USD 300,000,000 |
| Coupon rate & frequency | Interest schedule — fixed, floating, or zero | 3.2% fixed, semi-annual |
| Seniority | Ranking vs other issuer debt in liquidation | Unsecured & unsubordinated |
| Contingency provisions | Embedded options: call, put, or conversion | None (plain vanilla) |

**ANNUAL COUPON = PAR × COUPON RATE**

USD 300m × 3.2% = USD 9.6m per year, paid semi-annually as USD 4.8m. At maturity, the investor receives the final USD 4.8m coupon PLUS the USD 300m principal for a total USD 304.8m.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 1, "Features of Fixed-Income Securities", pp. 7–12.*

# Coupons come in three flavours — fixed, floating, or zero — and a floater's rate is always MRR plus a fixed issuer spread

*The credit spread is set at issuance and stays constant; only the MRR moves — so an FRN's coupon tracks short-term rates but keeps a fixed credit cushion.*

```
FRN coupon  =  MRR  +  Credit spread          Quarterly interest
                    =  (Coupon × Principal) / 4
```

*where: MRR = market reference rate (e.g. SOFR, €STR); Credit spread is issuer-specific and fixed at issuance.*

## CFA LM 1 — ANTELAS AG FOUR-YEAR FRN

EUR 250m principal · coupon = 3-month MRR + 250 bps · paid quarterly · MRR for this quarter = 0.25%.

**1.** Coupon rate this quarter = 0.25% MRR + 2.50% spread = 2.75% p.a.
**2.** Annual interest = 250,000,000 × 2.75% = EUR 6,875,000
**3.** Quarterly interest = 6,875,000 / 4 = EUR 1,718,750
**4.** If MRR moves to 0.50% next quarter, coupon resets to 3.00% → EUR 1,875,000 — spread stays put at 250 bps.

**Quarterly payment = EUR 1,718,750 · coupon moves with MRR · credit spread fixed at 250 bps**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 1, p. 9 (FRN); LM 2 Example on Antelas, p. 33.*

# The indenture is the legal contract; covenants split into affirmative (must-do) and negative (must-not-do), and tighten sharply as credit quality falls

*BRWA is investment grade and gets a short covenant list. VIVU is high-yield secured and wears a full monitoring regime — the covenant stack is the credit premium made legal.*

### BRWA · INVESTMENT GRADE · UNSECURED

AFFIRMATIVE covenants only:

- Use-of-proceeds defined
- Pari passu with other unsecured debt
- Timely financial reporting

Light NEGATIVE covenants:

- Limitation on liens
- Limitation on sale & leaseback
- Merger/consolidation protection

No leverage tests. No dividend blocks. No call.

### VIVU · HIGH YIELD · SECURED CALLABLE

Above + security pledge PLUS:

- Restriction on additional debt
- Dividend / distribution block
- Debt/EBITDA ≤ 5.00×
- Interest coverage ≥ 2.50×

INCURRENCE TEST (tighter 4.5× / 3.0×) must be met before paying dividends or issuing new debt.

Plus fixed-price call schedule — issuer retains refinancing flexibility.

Covenant violations → increased interest rate, accelerated payment, or termination of debt agreement. Strict covenants can also force default — not always in bondholders' interest.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 1, "Bond Indentures and Covenants", pp. 14–19.*

# Callable bonds let the issuer buy bonds back at a predetermined schedule — cheap refinancing for the issuer, call risk for the investor

*The call price declines toward par as maturity approaches — VIVU pays a premium for the flexibility, and that premium shrinks as optionality becomes less valuable.*

| Window | Call price (% par) |
|---|---|
| Years 0–3 (protection) | Non-callable |
| Years 3–4 | 103.25% |
| Years 4–5 | 102.50% |
| Years 5–6 | 101.75% |
| Years 6–7 | 101.00% |
| At maturity | 100.00% |

**SOLUTION — WHEN DOES VIVU CALL?**

VIVU 6.5% 7y callable, USD 400m. If yields fall 200 bps 3 years in, VIVU can refinance at 4.5%.

**1.** After call protection ends (yr 3) → call price is 103.25% = USD 413m
**2.** Annual interest saving on refi: 400 × (6.5% − 4.5%) = USD 8m
**3.** Call premium paid: USD 13m — recouped in <2 years
**4.** Economics work → VIVU calls. Investor is forced to reinvest at the lower 4.5% rate.
**5.** Callable bond YTM > equivalent non-callable — investor is compensated for call risk ex ante.

**Issuer refinances cheaply; investor faces reinvestment risk + capped upside**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, Exhibit 9 (VIVU), pp. 39–42.*

SECTION 02

# Cash-flow structures

# Bullet, amortizing, and balloon bonds differ entirely in when principal comes back to the investor — the interest is mechanical once that choice is made

*Same 5-year, USD 300m par, 3.2% semi-annual coupon — three completely different cash-flow profiles for the investor.*

| Year | Bullet (USD m) | Partially amortizing — 150m balloon | Fully amortizing |
|---|---|---|---|
| 0 | −300.0 | −300.0 | −300.0 |
| 0.5 | +4.8 | +18.75 | +32.7 |
| 1.0 – 4.5 (each period) | +4.8 | +18.75 | +32.7 |
| 5.0 (maturity) | +304.8 | +168.75 | +32.7 |
| **Total principal** | **At maturity** | **Spread + 150m balloon** | **Entirely over life** |

**BULLET**

Standard corporate bond. Interest-only coupons, principal in one lump at maturity. Highest reinvestment risk for investor.

**FULLY AMORTIZING**

Each payment = interest + principal. Level cash flow. Lower credit risk, higher reinvestment risk. Typical for mortgage loans.

**PARTIAL + BALLOON**

Hybrid: partial amortization plus a balloon at maturity. CRE loans often use this structure.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, "Fixed-Income Cash Flow Structures", pp. 27–31.*

# A fully-amortizing payment is just an annuity solved for PMT — early payments are mostly interest, late payments are mostly principal

*Same identity as a PV-of-annuity rearranged: given the loan size, the rate, and the number of periods, solve for the level payment.*

$$A = r \times \text{Principal} \;/\; [\,1 - (1 + r)^{(-N)}\,]$$

*where: A = periodic payment; r = rate per period (monthly here); Principal = loan amount; N = number of payments.*

**CFA LM 2 — USD 400,000 MORTGAGE · 30Y · 3.50% NOMINAL**

Monthly rate = 3.50% / 12 = 0.29167%; N = 360 months.

**1.** r × P = 0.0029167 × 400,000 = 1,166.67
**2.** Denominator = 1 − (1.0029167)^(−360) = 1 − 0.3499 = 0.6501
**3.** Monthly payment A = 1,166.67 / 0.6501 × (1 / 0.0029167 adj) = USD 1,796.18
**4.** Total paid over 30 years = 1,796.18 × 360 = USD 646,625

**USD 1,796.18 / month · USD 246,625 interest on USD 400,000 principal**

| Month | Interest | Principal | Remaining |
|---|---|---|---|
| 1 | 1,166.67 | 629.51 | 399,370 |
| 2 | 1,164.83 | 631.35 | 398,739 |
| 3 | 1,162.99 | 633.19 | 398,106 |
| … | ... | ... | ... |
| 358 | 15.63 | 1,780.55 | 3,577 |
| 359 | 10.43 | 1,785.75 | 1,791 |
| 360 | 5.22 | 1,790.96 | 0 |

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, Example 1 — 30-year Fixed-Rate Residential Mortgage, pp. 31–32.*

# Sinking funds retire principal through the life of the bond; ABS waterfalls route cash flows through tranches A→B→C in priority order

*Both structures reduce credit risk for senior investors at the cost of reinvestment risk — but through very different mechanisms.*

## SINKING FUND

Issuer sets aside cash periodically into an escrow account to retire a predetermined amount of principal early.

Mechanics:

- Trustee selects bonds by random lottery
- Or issuer repurchases at a fixed sinking-fund price
- Principal outstanding declines each year before maturity

Effect: credit risk falls (smaller outstanding balance), reinvestment risk rises.

## WATERFALL (ABS / MBS)

Cash flows from underlying loans route sequentially through tranches:

1. Tranche A (senior) gets principal FIRST
2. Tranche B gets principal ONLY after A fully repaid
3. Tranche C (junior) gets principal ONLY after A & B fully repaid

Payment shortfalls absorbed bottom-up: Tranche C loses first, Tranche A last. Typical in RMBS, CMBS, CLO structures.

Tranche A spread < Tranche B spread < Tranche C spread — higher coupon compensates for higher loss priority.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, pp. 32–33.*

# Floating-rate, step-up, payment-in-kind, and inflation-linked bonds are all ways to make the coupon stream adaptive rather than fixed

*Each mechanism shifts a specific risk — interest rate, ESG, credit deterioration, or inflation — between the issuer and the investor.*

| Structure | Coupon mechanism | Typical use case |
|---|---|---|
| Floating-rate note (FRN) | MRR + fixed spread; resets each period with MRR | Bank-funded loans |
| Step-up bond | Coupon rises by a preset schedule (or on credit-rating trigger) | ESG-linked bonds |
| Credit-linked coupon | Coupon adjusts with leverage / ICR (Antelas: 250→325 bps by TLR) | Leveraged loans |
| Payment-in-kind (PIK) | Interest paid as new principal — no cash outflow | Highly levered LBOs |
| Index-linked (TIPS) | Principal adjusted by CPI each period; coupon = rate × | Sovereign |

**CFA LM 2 EXAMPLE 2 — TIPS CASH FLOW**

USD 115m principal · 1.00% semi-annual coupon · CPI rises 265.00 → 267.25 (+0.849%)

**Adjusted principal = 115.976m · Semi-annual coupon = 1.00% × 115.976m / 2 = USD 579,882**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, pp. 33–37 (incl. Antelas leveraged loan, PPC SLB, TIPS Example 2).*

# Zero-coupon bonds replace coupons with a single discount at issuance; deferred-coupon bonds simply push coupons back in time

*Both pay no periodic interest early on — but their motives, pricing, and risk profiles are different animals.*

## ZERO-COUPON

Single cash flow: par at maturity. Priced below par if $r > 0$ — the implied interest is the discount.

Why investors love them:

- No coupon-reinvestment risk
- Perfect duration matching for fixed future obligations
- Useful for cash-flow matching in pension/insurance ALM

Sources:

- Sovereigns at short tenors (T-bills)
- Stripped from coupon-bearing sovereigns ("STRIPS")
- Long-dated corporate zeros post-2008

## DEFERRED-COUPON

Zero coupons for the first few years, then higher coupons through maturity. Extreme case = zero-coupon itself.

Why issuers use them:

- Conserve cash near issuance
- Common where project does not generate income until completion (e.g., construction)
- Signal of lower credit quality

Pricing:

- Usually at a discount to par
- Higher later coupons do not fully offset zero-coupon years

Both are priced today as the PV of known future cash flows — the same bond-math framework from Deck 01.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, pp. 37–38.*

# Put and conversion provisions hand optionality to the bondholder — puts provide a price floor, conversion trades debt for equity

*These embedded options are priced in at issuance — putable bonds yield less than option-free; convertibles yield even less because conversion adds equity upside.*

## CFA LM 2 — ZTG BIOTECH FIVE-YEAR CONVERTIBLE

EUR 300m convertible; 1.25% coupon; Face EUR 1,000 per unit; current share EUR 28; conversion price EUR 42.

**1.** Conversion ratio = Face / Conversion price = 1,000 / 42 = 23.81 shares per bond unit
**2.** Conversion premium at issuance = (42 − 28) / 28 = +50.0% over current share price
**3.** One year later share price at EUR 35 → Conversion value = 23.81 × 35 = EUR 833.35
**4.** Bond trades at ~EUR 1,000 (par) → investor holds on, collects 1.25% coupon, waits for share to rise above EUR 42.

**Convert only when share price > conversion price (EUR 42)**

## PUTABLE BOND

Investor can sell bond back to issuer at par if rates rise → put price is a floor. Issuer pays lower coupon for the privilege.

## CONTINGENT CONVERTIBLE (CoCo)

Bank capital instrument. Automatically converts to equity on trigger (e.g., CET1 ratio breach). Higher yield to compensate for involuntary conversion.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, Exhibit 10 (ZTG Biotech), pp. 40–43.*

SECTION 03

# Issuance & trading

# Fixed-income markets are sliced along three axes at once — issuer type, credit quality, and time to maturity — and every instrument lives in a specific cell

*Each cell has a natural issuer and a natural investor. Match mismatches drive alpha — and drive blow-ups when short-term funding meets long-term assets.*

| Credit quality | Short-term (<1y) | Intermediate (1-10y) | Long-term (>10y) |
|---|---|---|---|
| "Default risk-free" (DM sov) | Treasury bills | Treasury notes | Treasury bonds |
| Investment grade corp | Commercial paper · repo · ABCP | Unsecured corporate bonds · ABS | Unsecured corporate bonds · MBS |
| High yield corp | (limited access) | Secured corporate bonds · leveraged loans | Secured corporate bonds |

## NATURAL INVESTORS BY CELL

Money-market funds → short-term risk-free and IG. Financial intermediaries & central banks → intermediate risk-free. Bond funds & ETFs → intermediate IG. Pension funds & insurers → long-term risk-free and IG (liability matching). Asset managers → long IG. Hedge funds & distressed-debt funds → HY and special situations.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, "Fixed-Income Segments, Issuers, Investors", pp. 57–63.*

# Credit ratings create a bright line at BBB-/Baa3 — below it, "high yield" or "junk"; above it, "investment grade"

*Many institutional portfolios are hard-capped at investment-grade ratings — the line is regulatory, contractual, and psychological, all at once.*

| Tier | S&P rating | Meaning | Market share / role |
|---|---|---|---|
| Prime / gilt-edged | AAA | Extremely strong capacity to meet obligations | Sovereigns, top supranationals |
| Upper IG | AA / A | Very strong / strong; some sensitivity to adverse conditions | Large IG corporates |
| Lower IG | BBB+ / BBB / BBB- | Adequate; more vulnerable to downturns | Mainstream IG bond indices |
| Highest HY | BB+ / BB / B | Uncertainty in meeting obligations; near-term stability OK | Leveraged finance market |
| Deep HY / distressed | CCC / CC / C | Currently vulnerable; depends on favorable conditions | Hedge funds, distressed |

**FALLEN ANGEL**

Former IG issuer downgraded to HY. Outstanding debt carries IG features (long maturity, few covenants, non-callable) — but investors must dump it, pressuring price as HY market is a fraction of IG size.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, S&P Rating Scale, pp. 59–61.*

# Fixed-income indexes have 10×+ more constituents and 10×+ more turnover than equity indexes — market-value weighted, monthly-rebalanced

*Because bonds mature and issuers refinance, bond indexes churn constantly — funds tracking them hold a representative sample, never the full basket.*

### BLOOMBERG GLOBAL AGGREGATE

Scope: fixed-rate IG bonds from sovereign, corporate, securitized issuers in 28 developed & emerging markets.

Filters:
- IG rating
- ≥ 1y to maturity
- Min issue sizes by market

Excludes: HY, unrated, sub-min-size, below-min-maturity.

Monthly rebalance.

### JPM EMBI+ (EM SOVEREIGN)

Scope: USD-denominated EM sovereign debt.

Filters:
- Sovereign rated Baa1 / BBB+ or below
- USD-denom only (no FX-linked)
- Min USD 500m size
- ≥ 2.5y to maturity

Purpose: dollar-based return premium over DM sovereigns in exchange for EM credit + political risk.

Monthly rebalance.

### BLOOMBERG MSCI EUR CORP SRI

Scope: EUR-denominated IG corporate bonds with ESG overlay.

Filters:
- IG rating (≥ BBB-)
- MSCI ESG rating ≥ BBB
- EUR 300m min size
- ≥ 1y maturity

Exclusions: alcohol, tobacco, gambling, adult entertainment, GMOs, nuclear, firearms, weapons, thermal coal, oil sands, MSCI "Red" controversy score.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, "Fixed-Income Indexes", pp. 65–69.*

# Investment-grade corporate issuance compresses into a single trading day — announce at 9:15, price at 3pm, free to trade by next morning

*Frequent IG issuers operate under a shelf registration and pick the day opportunistically — the entire "roadshow" collapses to a 90-minute investor call.*

| Time | Event | Market state |
|---|---|---|
| 9:00 am | Underwriters and issuer agree to launch transaction | Pre-announcement |
| 9:15 am | Transaction announced – size, spread, maturities still TBD | Order book opens |
| 10:30 am | Investor conference call with issuer and underwriters | Order book building |
| 1:00 pm | Price guidance published; size and maturities firm up | Indications refined |
| 1:00–3:00 pm | Order book closed; underwriters allocate to institutional investors | Allocation |
| 3:00 pm | Transaction priced; government-benchmark spread and coupon fixed | Priced |
| 4:00 pm | Final term sheet delivered; bonds free to trade same or next day | Secondary |

**CONTRAST — HY & SECURED**

HY / secured issuance takes weeks: complex covenants, collateral diligence, best-efforts (not underwritten) distribution. Private placements skip the public process entirely for <~30 investors.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, Exhibit 6 (IG Corporate Issuance Timeline), pp. 70–72.*

# Secondary bond markets are OTC and quote-driven; liquidity is concentrated in on-the-run sovereigns, and seasoned corporates trade on 10–20bp spreads

*Unlike equities, bonds rarely trade on exchanges — price discovery lives at dealer desks, and bid-offer is the honest measure of how liquid a line really is.*

| Bond type | Typical bid-offer | Why |
|---|---|---|
| On-the-run DM sovereign (e.g. latest UST 10y) | < 1 bp | Primary dealers make deep markets |
| Recent IG corporate from frequent issuer | A few bp | Dealer inventory; index demand |
| Seasoned IG corporate / infrequent issuer | 10–20 bp+ | Thin dealer inventory |
| High-yield corporate | 25–75 bp | Smaller market; wider dispersion |
| Distressed debt | > 100 bp, deep discount | Specialized hedge-fund buyers |

**HERTZ 2020 — HOW DISTRESS LOOKS IN THE TAPE**

May 2020: Hertz filed Chapter 11 as COVID wiped out car-rental cash flows. 2022 and 2028 Hertz bonds traded < 10 cents on the dollar — a > 90% drop in 3 months. One year later those same bonds recovered to pre-pandemic levels after a restructuring left them intact. Meanwhile Hertz equity had been delisted months earlier.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, "Secondary Fixed-Income Markets", pp. 72–74.*

SECTION 04

# Corporate FI markets

# Corporate short-term funding ranges from cheap-and-uncommitted bank lines up to firm multi-year revolvers with full covenants — reliability trades against cost

*As you climb the reliability ladder, the bank asks for upfront commitment fees and covenants in exchange for legally binding liquidity.*

| Facility | Reliability | Bank commitment | Cost to borrower | Covenants |
|---|---|---|---|---|
| Uncommitted line of credit | Lowest | Revocable at bank's discretion | MRR + spread on drawn only | None |
| Committed line (364-day) | Medium | Formal written commitment | + commitment fee (e.g. 50bp) | Light |
| Revolving credit ("revolver") | Highest | Multi-year binding commitment | + commitment fee + covenants | Full set |
| Secured loan / asset-based | Credit-dependent | Backed by pledged collateral | Lower rate vs unsecured | Per collateral |
| Factoring of receivables | n/a | Sale of receivables to factor | Discount on face value | None |

**THE RULE OF THUMB**

Strong IG borrowers with stable deposits at the bank prefer uncommitted lines (no fee). HY borrowers need revolvers (binding liquidity + covenant monitoring). Revolvers typically backstop a commercial-paper programme.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "Short-Term Funding Alternatives", pp. 80–82.*

# Commercial paper is unsecured, <90-day, and rolled over constantly — the rollover risk is plugged by a committed backup liquidity line

*About 60% of annual CP issuance is by financial institutions; ABCP wraps short-term loans into an SPE and keeps the loans off the bank's balance sheet.*

## CP (UNSECURED)

Issued by IG corporates and financial institutions.

- Usually < 3 months maturity
- Unsecured promissory note
- Rolled over at maturity with newly issued CP
- Rollover risk plugged by a committed backup liquidity line from banks
- Eurocommercial paper = USCP's international cousin, smaller trade size, less liquid

## ABCP (SECURED)

Bank transfers short-term loans / receivables to an SPE.

- SPE issues CP backed by those assets
- Bank provides a backup liquidity line (not a guarantee)
- Off-balance-sheet for the bank → reduced capital charge
- Investors get exposure to a loan pool they could not access directly
- Nearly collapsed 2007–2008 under rollover pressure

## ALSO — INTERBANK & CDs

Financial institutions borrow from each other overnight to ~1y at rates tied to the MRR.

Certificates of deposit:

- < 1 year, pay interest at maturity
- Non-negotiable → early-withdrawal penalty
- Negotiable → tradeable pre-maturity
- Large-denom CDs = major wholesale bank funding source

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "External, Security-Based Financing" and ABCP, pp. 82–85.*

# A repurchase agreement is a collateralised loan dressed as a sale — the repo rate is the interest, the haircut is the overcollateralisation

*Legally a sale + forward repurchase; economically a secured loan. Initial margin > 100% means the cash lender is overcollateralised; the haircut is the borrower's skin in the game.*

```
Initial margin  =  Security price / Purchase price        Haircut  =  (Sec px –
 Purchase px) / Sec px        Repurchase px  =  Purchase px × [1 + r × d/360]
```

*where: r = repo rate (annualized); d = days to repo maturity; variation margin = (Initial margin × Purchase $px_t$) − Security $px_t$, transferred to restore original margin.*

**CFA LM 4 — 30-DAY REPO ON USD 100m US 5y TREASURY · REPO RATE 0.25% · INITIAL MARGIN 102%**

Seller (cash borrower) delivers USD 100m of Treasuries as collateral to buyer (cash lender).

**1.** Purchase price (loan amount) = Security price / Initial margin = 100,000,000 / 1.02 = USD 98,039,216
**2.** Haircut = (100,000,000 − 98,039,216) / 100,000,000 = 1.96% — seller funds remaining 1,960,784 elsewhere
**3.** Repurchase price = 98,039,216 × [1 + (0.25% × 30 / 360)] = USD 98,059,641
**4.** Repo interest earned by cash lender = 98,059,641 − 98,039,216 = USD 20,425

**Lender lends 98.04m against 100m UST for 30 days → earns 20,425 of repo interest (haircut 1.96%)**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "Repurchase Agreements" Knowledge Check, pp. 86–89.*

# Reverse repos let hedge funds short bonds — borrow the security, sell it, wait for the price to fall, buy it back cheaper, return it

*From the cash lender's point of view the trade is a reverse repo — but the real motive may be renting the specific security to support a short.*

| Step | Time | Hedge-fund action | Cash flow / P&L |
|---|---|---|---|
| 1 | t = 0 | Borrow USD 100m 5y UST in repo market; deliver USD 100m cash | −100m cash, +UST |
| 2 | t = 0 | Simultaneously sell the UST in the cash market at par | +100m cash, −UST |
| 3 | t = T | Buy back the UST in the cash market at 99% of face | −99m cash, +UST |
| 4 | t = T | Deliver UST back to repo counterparty, receive repurchase price USD 100.02m | +100.02m cash, −UST |
| **Net** | **Over 30 days** | **Short works if price falls more than repo cost** | **Profit USD 1.02m** |

## SPECIAL COLLATERAL RATES — WHEN SHORT DEMAND MEETS SUPPLY

When one specific security (e.g. the latest on-the-run UST) is in heavy short demand, the repo rate on that specific collateral can fall sharply below the general-collateral rate — sometimes even below zero. A negative repo rate means the cash lender PAYS the borrower for the privilege of renting out a scarce security.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, Exhibit 5 + Knowledge Check, pp. 90–91.*

# Repos caused Bear Stearns to fail — USD 18bn cash became USD 2bn in two days when counterparties refused to roll

*The lesson is structural: repo is cheap for a reason — it's the first funding source to vanish when the firm becomes untrustworthy, and the collateralised nature does not save you.*

| Date | Event | Signal |
|---|---|---|
| End-2006 | Bear Stearns: USD 350bn assets, USD 9bn revenue — 5th-largest US broker-dealer | Healthy |
| 2007 | Subprime MBS market deteriorates; two Bear hedge funds fail in July | Amber |
| End-2007 | First quarterly loss ever; USD 2bn MBS writedown; credit downgrade | Amber |
| 2006→2007 | CP cut from USD 20.7bn → USD 3.9bn; shift into repo funding | Amber |
| 10 March 2008 | USD 18bn cash position | Red |
| 12 March 2008 | Cash down to USD 2bn — repo counterparties refuse to roll; prime-broker runs | Red |
| 14 March 2008 | Fed emergency loan against securities collateral | Rescue |
| 17 March 2008 | Bear Stearns sold to JPMorgan (USD 10/share final) with Fed backstop | Failed |

Repos are uncommitted: the cash lender has no obligation to roll. Loss of market confidence + short-term funding = instant liquidity crisis.

[2] Post-crisis reforms tightened triparty repo clearing, intraday margining, and Fed standing repo facility to reduce systemic vulnerability.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "The Fall of Bear Stearns" box, pp. 93–94.*

# Longer maturities mean higher rates AND higher spreads for the same issuer — the credit premium itself widens with time

*Investor and issuer sit on opposite sides of the same trade-off. The investor faces price and reinvestment risk; the issuer faces rollover risk and refinancing cost.*

| Maturity | BRWA yield | Government yield | Credit spread |
|---|---|---|---|
| 3-year | ~2.4% | 1.5% | ~90 bps |
| 5-year | 3.20% | 2.30% | 90 bps |
| 7-year | ~3.7% | ~2.6% | ~110 bps |

## INVESTOR VIEW

Extending maturity beyond horizon → price risk (must sell before maturity at uncertain price).

Shortening maturity below horizon → reinvestment risk (coupons and principal reinvest at unknown future rates).

## ISSUER VIEW

Short debt for a long project → rollover risk (refinance at higher rates).

Long debt for short project → locked-in high rate; prepayment penalties; suboptimal capital structure.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "Long-Term Corporate Debt", Exhibit 7 (BRWA), pp. 95–96.*

# Investment-grade bonds ride the government benchmark with thin spreads and minimal covenants; high-yield bonds wear equity-like cash flows and tight leash of covenants

*The YTM gap isn't just "more default risk" — it reflects a completely different contractual, analytical, and behavioural regime.*

| Dimension | Investment grade (BRWA) | High yield (VIVU) |
|---|---|---|
| YTM composition | Mostly government benchmark; thin credit spread | Benchmark + wide issuer-specific spread |
| Cash flow profile | Bond-like — high probability of scheduled payments | Equity-like — payment uncertain, recovery analysis |
| Analytical approach | Financial ratios, credit-rating migration | PD × LGD; covenant tests; collateral value |
| Covenant stack | Light (pari passu, liens, M&A limits) | Heavy (leverage, coverage, dividend blocks) |
| Maturity range available | Up to 30 years | Typically ≤ 10 years |
| Typical contingency features | Non-callable or make-whole call only | Fixed-price callable — issuer retains flexibility |
| Market access | Always open; opportunistic refi | Cyclical; restructuring / renegotiation needed |

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "Long-Term Corporate Debt", Exhibits 8–10, pp. 96–101.*

SECTION 05

# Government issuers

# A sovereign's economic balance sheet is expected tax revenues vs promised expenditures — not depreciation and accruals; fiscal policy sets level, debt-management sets composition

*The "default-risk-free" label is the endpoint of two facts: the sovereign can tax the entire economy, and it can print its own currency (for domestic-currency debt).*

## ECONOMIC BALANCE SHEET

CLAIMS (assets)
- Short-term & long-term financial claims
- Fixed assets — infrastructure, land, SOEs
- FX + commodity reserves
- PV of expected future tax revenues

OBLIGATIONS (liabilities)
- Currency outstanding
- Short- and long-term debt (domestic & FX)
- Pension & other domestic obligations
- Supranational / external obligations
- PV of promised future expenditures

## FISCAL vs DEBT-MANAGEMENT

FISCAL POLICY — sets the LEVEL of debt.
- Budget deficits → more borrowing
- Budget surpluses → less
- Sensitivity to growth and inflation forecasts

DEBT-MANAGEMENT POLICY — sets the COMPOSITION.
- Short vs long maturity
- Fixed vs floating vs linker
- Domestic vs foreign currency

OECD data (2021):
- Weighted avg maturity ≈ 7.6y (vs 6.3y 2007)
- Long-term debt ≈ 86% of total outstanding
- Annual issuance ~50/50 short vs long

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, "Sovereign Debt", pp. 107–113.*

# Developed-market sovereigns issue "default-risk-free" debt with unconstrained maturity access; emerging-market sovereigns face currency restrictions and external-debt stress

*EM external debt is particularly fragile: investors face indirect FX risk because the issuer needs foreign-currency revenues to service the debt.*

| Dimension | Developed-market sovereign | Emerging-market sovereign |
|---|---|---|
| Currency of issuance | Reserve currency (USD, EUR, JPY, GBP, CHF, AUD, CNY) | Restricted domestic currency + external USD/EUR |
| Maturity access | Unconstrained — full curve to 30y+ | Often capped by domestic-currency depth |
| Tax base | Broad, diversified, stable | Commodity-concentrated, cyclical |
| Default risk | Effectively zero for domestic-currency debt | Material, especially on external debt |
| Fiscal-policy transparency | High — steady rules, published accounts | Variable — higher sensitivity to politics |

**SRI LANKA EXTERNAL-DEBT DEFAULT — MAY 2022**

First Asia-Pacific default in many years. Govt debt/GDP jumped 87% → 100%+ by 2020 as infrastructure spending was financed by USD bonds, World Bank, ADB, and major trading partners (Japan, China). Tourism + remittances collapse + 2022 inflation/commodity spike → foreign-currency reserves ran out. Missed USD 78m coupon → 30-day cure → default. External bond creditor split: market bonds 47%, ADB 13%, Japan 10%, China 10%, World Bank 9%, India 2%, other 9%.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, pp. 108–110; Sri Lanka case study p. 109.*

# Sovereign debt is sold at public auction — single-price auctions give everyone the same yield; multiple-price auctions pay each winner their bid

*Single-price auctions reduce yield volatility and "winner's-curse" fear — which is why the RBI switched in 2021 after a series of partially failed auctions.*

| Dimension | Single-price (Dutch) auction | Multiple-price (discriminatory) |
|---|---|---|
| Price paid by winners | All winners pay the same — the stop/clearing price | Each winner pays their own bid |
| Ranking logic | Bids ranked low-yield → high-yield until size filled | Same ranking; different pricing |
| Yield volatility | Lower — no winner's curse | Higher dispersion of winning prices |
| Distribution of allocation | Broader — encourages aggressive bids | Narrower — large bidders win at bid |
| Non-competitive bids | Always accepted at clearing price | Always accepted at average price |
| Role of primary dealers | Required to bid competitively in all auctions | Same — required competitive participation |

**CFA LM 5 — MAS 30-YEAR SGS AUCTION · 21 SEP 2021**

**Coupon set at cut-off rounded DOWN to nearest 0.125% → 1.875%. Cut-off price 98.303 → all winning bidders pay this.**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, Exhibit 2 Singapore 30y + Reserve Bank of India box, pp. 115–117.*

# Primary dealers win the auction; central banks, foreign reserves, and regulated institutions buy "non-economically" in the secondary market — it's why Treasury yields sit below credit curves

*The non-economic bid is structurally large and persistent — it is part of what makes the "default-risk-free" yield genuinely low, and drives the liquidity premium of on-the-run sovereigns.*

| Buyer | Motive | Why "non-economic" |
|---|---|---|
| Central bank (domestic) | Conduct monetary policy; OMO, repo, QE | Policy target not return |
| Foreign central banks | Hold FX reserves in reserve-currency sovereigns | Reserve asset not trade |
| State & local governments | Investment-policy restrictions favour sovereigns | Regulatory constraint |
| Banks | Meet LCR / HQLA; capital charge 0% under Basel | Regulatory preference |
| Insurance companies | ALM liability matching; capital relief on sovereigns | Regulatory incentive |

## USD AS GLOBAL RESERVE CURRENCY

USD ≈ 60% of global FX reserves · ≈ 50% of international trade · ≈ 50% of global debt securities · ≈ 90% of FX-market transactions. EUR, JPY, GBP, CHF, AUD, CNY also serve as reserve currencies. This structural demand compresses US Treasury yields below where they would trade on pure credit/rate fundamentals — the "exorbitant privilege".

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, "Sovereign Debt Issuance and Trading", pp. 117–118.*

# Non-sovereign issuers split cleanly into three families — agencies backed by sovereigns, local authorities backed by taxes/revenues, and supranationals backed by member states

*All three sit between pure sovereign and pure corporate credit — with very different primary and secondary repayment sources.*

| Issuer type | Primary repayment source | Secondary source | Example |
|---|---|---|---|
| Government agency | Cash flows from agency operations / guaranty fees | Sovereign government backing | AAHK, Ginnie Mae |
| Local GO bond | Local tax revenues | Broader regional tax base | Province of Ontario |
| Local revenue bond | User fees / tolls from a specific project | Project-level reserves | Toll-road authority |
| Supranational | Member-state capital + loan repayments | Implicit/explicit state backing | World Bank, IMF, ADB |

## TWO WORKED EXAMPLES — ONTARIO GREEN BOND vs ADB GMTN

Province of Ontario 1.55% 2029 CAD 2.75bn Global Green Bond — GO bond, repaid from Ontario taxes, proceeds ring-fenced to eligible green projects (clean transport, renewables, etc.).

Asian Development Bank 7.80% 2034 Global MTN — supranational issued in IDR but payable in USD at spot FX on each payment date; investor assumes IDR risk in exchange for high IDR coupon.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, "Non-Sovereign, Quasi-Government, Supranational Agency Debt", pp. 118–122.*

KEY TAKEAWAYS

# Deck 13 — Fixed Income I — Instruments, Markets & Issuance

- Six bond features fully specify any bond: issuer, maturity, principal, coupon (fixed / floating / zero), seniority, and contingency provisions. Annual coupon = par × rate.
- Bullet, amortizing, and balloon differ only in when principal is returned. Fully amortizing payment = PMT of annuity; early payments mostly interest, later mostly principal.
- Credit-rating cut-off BBB-/Baa3 is the legal + contractual line between IG and HY. Fallen angels get dumped because institutional mandates force it.
- Sovereign auctions: single-price gives all winners the same yield (Singapore cut-off 1.95% → coupon 1.875%); multiple-price pays each winner their bid. RBI switched to single-price in 2021.
- FRN coupon = MRR + fixed issuer spread. Antelas EUR 250m @ MRR +250bps, MRR 0.25% → quarterly interest EUR 1,718,750.
- Covenants tighten as credit quality falls: BRWA (IG) has light affirmative and 3 negatives; VIVU (HY) adds leverage tests, dividend blocks, and a fixed-price call schedule.
- Repo math: Initial margin 102% on USD 100m UST → purchase price 98.04m (haircut 1.96%); 30-day repo at 0.25% → repurchase 98.06m. Bear Stearns died because counterparties refused to roll.
- Non-economic demand (central banks, foreign reserves, regulated institutions) compresses sovereign yields — the USD "exorbitant privilege" is structural, not transient.